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Department of Applied Chemistry (2024-25)

NOTES

UNIT-III

ENERGY STORAGE DEVICES

Branch: CV, ME, EE, EL, ET, VLSI

Unit 3

ENERGY STORAGE DEVICES

1. Mapped CO-2: **Apply** concepts of electrochemistry for energy storage devices and corrosion. (L3)
   Mapped PO – PO1, PO2, PO3, PO7, PO10, and PO12.

NOTES

SYLLABUS: Battery: Introduction, Characteristics, and General applications Lithium-ion battery, Glass battery, H_2-O_2 Fuel cell. Differences between Battery and Fuel cell. Recycling and safe disposal of batteries. Supercapacitors: Definition, Types, Characteristics, and Application. H2 as a green fuel: Introduction, Production, Storage, and Utilization.

Introduction:
Batteries are the most used energy storage devices. They are considered as storehouses for electrical energy on demand. Batteries are electrochemical devices which convert chemical energy into electrical energy. They use chemical reactions to produce electricity. Such electrochemical devices are called galvanic cells in contrast to electrolytic cells which consume electrical energy to bring about a chemical reaction.

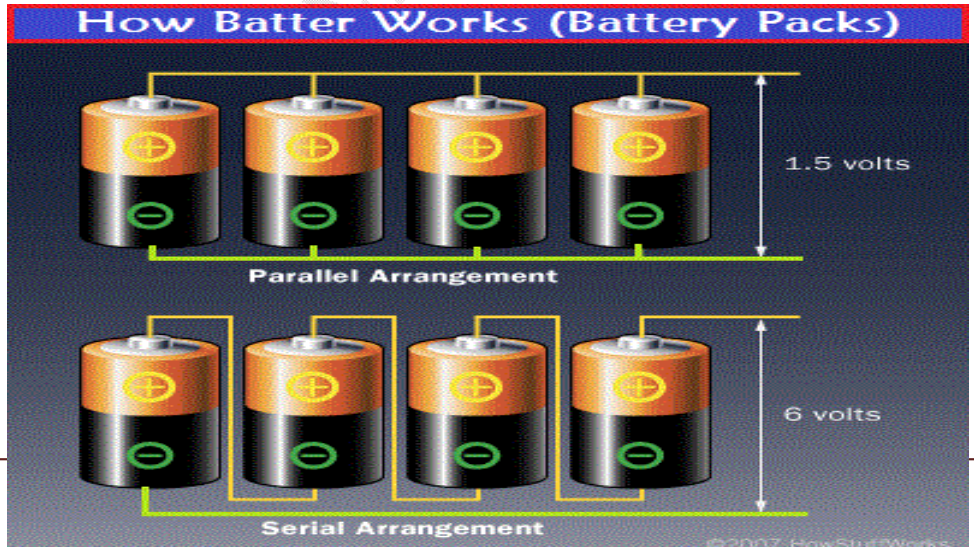
A battery consists of two electrodes, an anode and a cathode, and an electrolyte through which electrically charged particles can move. Two chemical reactions take place at the same time. The reaction taking place at anode is an oxidation reaction that leads to an excess of electrons at the anode. It is also called the negative electrode. The chemical reaction taking place at the cathode is a reduction reaction which results in depletion of electrons. Therefore, the cathode is also called the positive electrode. For example, in nickel cadmium battery, C d is oxidized to C d at the anode and Ni is reduced to N i at the cathode and the reaction can be represented as:

$Cd \rightarrow Cd^{2+} + 2e^-$ at anode (- ve electrode)

$Ni^{3+} + e^- \longrightarrow Ni^{2+}$ at Cathode (+ve electrode)

When the battery is connected to a external circuit, excess electrons from the anode flow through the circuit and back to cathode. As the electrons move through the circuit they lose energy. This energy may be used to create heat or light as in an electrical heater or light bulb, or to do work as in a motor. The flow of electrons result in a current and by convention the direction of flow of current is taken as opposite to the direction of flow of electrons.. The energy release per unit charge while the current flow through the circuit is called voltage.

Parallel: batteries are arranged next to each other, with the external electric circuit connecting to each other, with the electric circuit connecting to every battery's anode and cathode terminal.
Series: batteries are arranged end to end, with the external electric circuit connecting to the first batteries anode and last batteries cathode. The voltage adds up and the current stays the same.



Types of Batteries:

A battery is a source of electrical energy, which is provided by one or more electrochemical cells of the battery after conversion of stored chemical energy. In today's life, batteries play an important part as many household and industrial appliances use batteries as their power source. Batteries can be divided into two major categories, primary batteries and secondary batteries.

1 Primary Battery: A primary battery is a disposable kind of battery. Once used, it cannot be recharged. Alkaline batteries, Mercury batteries, Silver-Oxide batteries, and Zinc carbon batteries are examples of primary batteries.

Advantages:

1. High energy density since no design compromises is necessary to accommodate recharging.
2. The best alternative for low-cost low drain applications such as watches or hearing aids.
3. The good choice for single use operations such as guided missiles and military ordnance.
4. Low initial cost

Disadvantages:

1.It is not suitable for high drain applications due to the short lifetime and cost of continuous replacement.
2.It also produces much more waste than rechargeable batteries.
3.In terms of overall energy efficiency, single use, disposable, primary batteries are very uneconomical energy sources since they produce only about 2% of the power used in their manufacture.

2. Secondary Battery: Secondary batteries are rechargeable batteries. Once empty, it can be recharged again. This charging and discharging can happen many times depending on the battery type.
It can be used like the primary battery, then recharged and used again, the cycle is repeated until the capacity fades or is lost suddenly due to internal short-circuit.

The criteria for good secondary batteries:

1. Have longer life, high power density.
2. It should not leak or vent matter from it.
3. It should have a low cost per unit energy.
4. The active materials in a battery should be stable.
5. It should be usable over a wide temperature.
6. It should maintain its capacity from cycle to cycle, and overcharge ability.
7. It should be safe and easily disposable.

Differentiate between Primary and Secondary Battery:

Sr. No.	Primary Battery	Secondary Battery
1	Chemical reactions are irreversible.	Chemical reactions are reversible.
2	Only the chemical energy is converted into electrical energy when the current is drawing from it.	Electrical energy is converted into chemical energy when charged or chemical to electrical when discharged.

3	It cannot be recharged.	It can be recharged.
4	Its internal resistance is high.	Its internal resistance is low.
5	Only weak current is drawn.	A heavy current can be drawn.
6	Light and cheap.	Heavy and costly.
7	Disposable	Regular maintenance required.
8	Higher life cycle cost.	Lower life cycle cost.

Important terms related to battery:

1. **Energy Density:** It is defined as the ratio of the energy available from a cell or battery to its volume. Units: watt hour/Kg

2. **Power density:** It is defined as the ratio of the power available from a cell or battery to its volume. Units: (watt/liter).

3. **Energy efficiency of a secondary battery:**

The energy efficiency of a battery is the ratio of the energy output (discharged) to the energy input (charged), typically expressed as a percentage.

% Energy efficiency = (Energy released on discharge/ energy required for charge) x100

4. **Cycle life:** The cycle life of a battery refers to the number of charge-discharge cycles a battery can withstand before its capacity drops to a specified percentage (usually 80%) of its initial rated capacity.
5. **Shelf life:** Shelf life of a battery refers to the period of time a battery remains usable and retains its specified capacity when stored under recommended conditions, without being subjected to charge or discharge cycles. For rechargeable batteries, this is defined as the period between recharge cycles.

2)Lithium-ion battery:

Principle:

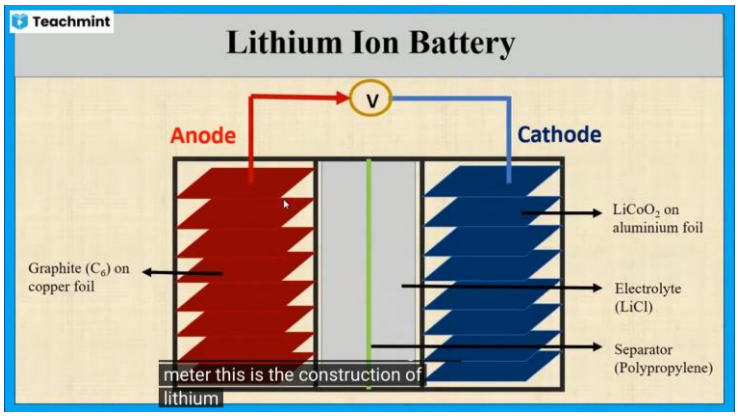
Lithium-ion batteries work on the principle of intercalation, where lithium ions move between the positive cathode and negative anode through an electrolyte.

Construction:

1. Positive Cathode: Lithium metal oxide (LiCoO2, LiNiCoAlO2)
2. Negative Anode: Graphite
3. Electrolyte: Lithium salts dissolved in organic solvents
4. Separator: Thin, porous membrane

Working:

1. Charging: Lithium ions move from cathode to anode through electrolyte.
2. Discharging: Lithium ions move from anode to cathode, generating electricity.



Cell representation:
Cu-C/LiCl// LiCoO2-Al

Discharge reaction:

At Anode: LixC → xLi+ + xe-+ C
At Cathode: Li (1-x) Co O2+ xLi+ + xe- → LiCoO2
Overall reaction: LixC+ Li (1-x) Co O2→ C+ LiCoO2

Recharge reaction:
At anode: LiCoO2 → Li (1-x) Co O2+ xLi+ + xe-
At cathode: C+ xLi+ + xe- → LixC
Electrolyte: Some salt of Lithium dissolved in organic solvent
Overall reaction: LiCoO2 + C → Li (1-x) Co O2 + LixC

Advantages:

1. High energy density (120-150 Wh/kg)
2. High discharge/charge efficiency (90-95%)
3. Long cycle life (300-500 cycles)
4. Low self-discharge rate (2-3% per month)
5. High power density
6. Compact design

Disadvantages:

1. High cost
2. Limited thermal stability
3. Flammability risks
4. Toxic materials (lithium, cobalt)
5. Aging effects
6. Recycling challenges

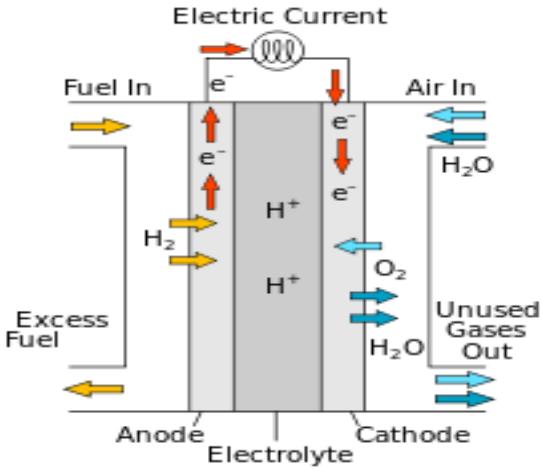
Applications:

1. Portable electronics (smartphones, laptops)
2. Electric vehicles (EVs)
3. Renewable energy systems (solar, wind)

4. Power tools
5. Medical devices
6. Aerospace and defense

In a lithium-ion battery the lithium ions are transported to and from the cathode or anode, With the transition metal, cobalt (Co),is being oxidized from Co3+ to Co4+ during charging, and reduced from Co4+ to Co3+ during discharge.

Fuel cell:
A H2-O2 Fuel Cell is an electrochemical device that converts chemical energy from hydrogen (H2) and oxygen (O2) into electrical energy, producing water and heat as by-products.



Reactions:
Cell: Pt, Fuel H_2 | KOH | O_2, Pt

Construction:

1. Anode (Negative Electrode): Hydrogen inlet, catalyst (e.g., Pt)
2. Cathode (Positive Electrode): Oxygen inlet, catalyst (e.g., Pt)
3. Electrolyte: Proton exchange membrane (PEM), solid polymer

Working:

1. Hydrogen fuel is fed to the anode.
2. Oxygen is fed to the cathode.
3. At the anode, hydrogen is oxidized, releasing electrons.
4. At the cathode, oxygen is reduced, accepting electrons.
5. Electrons flow through external circuit, generating electricity.
6. Protons (H+) migrate through electrolyte, combining with oxygen to form water.

Reactions:

Anode (Oxidation):
2H2 → 4H+ + 4e-

Cathode (Reduction):
O2 + 4H+ + 4e- → 2H2O

Overall Reaction:
$2H_2 + O_2 \rightarrow 2H_2O$ + Electricity + Heat

Advantages:

1. Zero greenhouse gas emissions
2. High efficiency (40-60%)
3. Low noise pollution
4. Reliability and durability
5. Flexibility in design and application
6. Abundant fuel sources (hydrogen)

Disadvantages:

1. High cost
2. Hydrogen production challenges
3. Storage and transportation issues
4. Limited infrastructure
5. Sensitivity to impurities
6. Durability concerns

Applications:

1. Transportation: Fuel cell electric vehicles (FCEVs)
2. Stationary Power: Backup power, combined heat and power (CHP)
3. Portable Power: Electronics, medical devices
4. Aerospace: Spacecraft, satellite power
5. Industrial: Power generation, chemical processing

Difference between Battery and Fuel cell

Sr. No.	Fuel cell	Battery
1	In fuel cell reactants are constantly supplied	In battery reactants are stored inside battery only.
2	A fuel cell is an open system where reactants are stored outside.	Battery is a closed system.
3	Fuel cells work as long as fuel is supplied.	Battery requires regular replacements.
4	Fuel cells require more space than reactants stored outside.	Battery requires less space.
5	Fuel cells can supply power for an indefinite length of time.	Battery supplies power for a definite length of time.
6	The amount of heat generated in fuel cells is much more.	The amount of heat generated in batteries is less.
7	By products must be constantly removed (Water) from fuel cells.	By products are not removed in case of batteries.
8	Fuel cells are more expensive.	Batteries are less expensive.

Glass battery:

Principle:

Glass batteries, also known as sodium-ion glass batteries or solid-state glass batteries use a glass electrolyte instead of liquid electrolytes found in traditional lithium-ion batteries. The glass electrolyte enhances safety, energy density, and charging speeds.

Working:

1. Sodium ions move through the glass electrolyte between cathode and anode.
2. Cathode: Sodium iron phosphate or other sodium-based materials.
3. Anode: Graphite or other carbon-based materials.(Sodium metal acts as the anode material).
4. Electrolyte: Sodium-ion conducting glass.
5. Separator: Thin glass layer separating cathode and anode.

Reactions:

Anode Reaction

$Na \rightarrow Na^+ + e^-$

Cathode Reaction

$FePO_4 + Na^+ + e^- \rightarrow NaFePO_4$

Overall Reaction:

Na (anode) + $FePO_4$ (cathode) → $NaFePO_4$ (sodium iron phosphate)

Advantages:

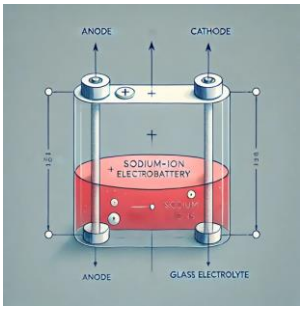
1. Improved Safety: Non-flammable glass electrolyte reduces fire risk.
2. Higher Energy Density: Glass batteries store more energy per unit.
3. Faster Charging: Glass electrolyte enables rapid ion conduction.
4. Long Cycle Life: Glass batteries withstand thousands of charge/discharge cycles.
5. Low Cost: Sodium replacement for lithium reduces material costs.
6. Environmentally Friendly: Non-toxic materials and recyclable.

Disadvantages:

1. Limited Scalability: Current manufacturing processes hinder mass production.
2. High Temperature Sensitivity: Glass electrolyte performance degrades at extreme temperatures.
3. Material Challenges: Glass electrolyte durability and stability issues.
4. Limited Compatibility: Incompatible with existing lithium-ion infrastructure.

Applications:

1. Electric Vehicles
2. Renewable Energy Systems
3. Consumer Electronics
4. Grid-Scale Energy Storage
5. Medical Devices



Recycling and safe disposal of battery:

Batteries run almost everything today, and it is impossible to get rid of them, the sustainability principle requires us to handle them in a safe and efficient manner to ensure that adverse environmental impacts can be avoided.

Recycling of Batteries:

Recycling of batteries is necessary because

1. To Conserve natural resources (e.g., lead, mercury, cadmium)
2. To Reduce waste and pollution
3. To Recover valuable materials (e.g., copper, nickel, cobalt)
4. To Decrease greenhouse gas emissions
5. To Protect public health and environment

There are different methods for recycling of batteries:

a) Mechanical Processing

1. Crushing: Breaking batteries into smaller pieces.
2. Shredding: Reducing battery pieces into fine particles.
3. Separation: Sorting materials by density, size, or magnetism.

b) Pyrometallurgical Processing

1. Smelting: Heating batteries to extract metals (e.g., lead, copper).
2. Refining: Purifying extracted metals.

c) Hydrometallurgical Processing

1. Leaching: Dissolving metals in aqueous solutions.
2. Precipitation: Separating metals from solutions.
3. Electrowinning: Depositing metals onto electrodes.

d) Electrochemical Processing

1. Electrolysis: Separating metals using electric current.
2. Electro winning: Depositing metals onto electrodes.

e) Chemical Processing

1. Solvent extraction: Separating metals using solvents.
2. Ion exchange: Removing impurities from solutions.

f) Biological Processing

1. Bioleaching: Using microorganisms to extract metals.
2. Bio precipitation: Removing metals from solutions using microorganisms.

Disposal of Batteries

Primary Disposal Methods:

1. Landfill Disposal: Batteries are sent to landfills, where they can leak toxic materials.
2. Incineration: Batteries are burned, releasing toxic fumes.
3. Recycling: Batteries are processed to recover valuable materials.

Different kinds of batteries require different methods of disposal. Some of the ways to deal with and dispose of the different kinds of batteries are mentioned as follows:

1. Lead-Acid (Pb) Batteries

- Used in: Automotive, industrial, and backup power applications
- Recycling method: Smelting and refining
- Recyclability: 99%
- Hazardous materials: Lead, sulfuric acid
- Disposal Method: Recycling (Mechanical separation), Pyrometallurgical separation

2. Nickel-Cadmium (Ni-Cd) Batteries

- Used in: Power tools, cordless phones, and other portable devices
- Recycling method: Pyrometallurgy
- Recyclability: 90%
- Hazardous materials: Cadmium, nickel
- Disposal Method: Recycling (chemical separation), Incineration

3. Nickel-Metal Hydride (NiMH) Batteries

- Used in: Hybrid and electric vehicles, cordless power tools, and other portable devices
- Recycling method: Hydrometallurgy
- Recyclability: 80%
- Hazardous materials: Nickel, rare earth metals
- Disposal Method: Recycling (Mechanical separation), Incineration

4. Lithium-Ion (Li-ion) Batteries

- Used in: Portable electronics, electric vehicles, and renewable energy systems
- Recycling method: Mechanical separation, electrochemistry, or hydrometallurgy
- Recyclability: 50-70%
- Hazardous materials: Lithium, cobalt, nickel
- Disposal Method: Recycling (electrochemical separation), Mechanical separation, Pyrometallurgical separation

5. Alkaline Batteries

- Used in: Flashlights, toys, and other household devices
- Recycling method: Mechanical separation and refining
- Recyclability: 70-90%
- Hazardous materials: None

- Disposal Methods: Dispose of in regular trash (check local regulations), Landfill Disposal, Recycling (mechanical separation)

6. Mercury-Containing Batteries

- Used in: Thermometers, fluorescent lighting, and other applications
- Recycling method: Distillation and refining
- Recyclability: 90%
- Hazardous materials: Mercury
- Disposal precautions: Participating recycling programs, do not incinerate or landfill

7. Silver-Oxide Batteries

- Used in: Watches, hearing aids, and other small devices
- Recycling method: Electrochemical refining
- Recyclability: 80%
- Hazardous materials: Silver
- Disposal Method: Recycling

8. Zinc-Carbon Batteries

- Used in: Flashlights, toys, and other household devices
- Recycling method: Mechanical separation and refining
- Recyclability: 70%
- Hazardous materials: None
- Disposal Method: Landfill Disposal, Recycling (mechanical separation)

Supercapacitors:

Definition:

Supercapacitors, also known as Ultra capacitors or Electrochemical Capacitors, store electrical energy through electrostatic double-layer capacitance and electrochemical pseudo capacitance.

Types:

1. Electrostatic Double-Layer Capacitors (EDLCs)
2. Electrochemical Pseudocapacitors (EPCs)
3. Hybrid Supercapacitors (combining EDLC and EPC)

1. **Electrostatic Double-Layer Capacitors (EDLCs)**

Electrostatic Double Layer Capacitors (EDLCs), also known as Supercapacitors or ultra-capacitors, store electrical energy through electrostatic double-layer capacitance and use activated carbon or carbon-based materials.

Characteristics: High power density, long cycle life, and low self-discharge

Applications: Power quality, UPS, renewable energy systems

2. Electrochemical Pseudocapacitors (EPCs)

Electrochemical Pseudocapacitors also known as Supercapacitors or ultra-capacitors, store electrical energy through electrochemical pseudo capacitance and use
Use metal oxides (RuO2, MnO2) or conducting polymers.

Characteristics: Higher energy density than EDLCs, but lower power density

Applications: Electric vehicles, hybrid electric vehicles

3. Hybrid Supercapacitors

Hybrid Supercapacitors also known as Supercapacitors or ultra-capacitors, store electrical energy through combine electrostatic double-layer capacitance and electrochemical pseudo capacitance. and use combination of activated carbon and metal oxides

Characteristics: Balance between power density and energy density

Applications: Electric vehicles, renewable energy systems

Advantages:

1. High Power Density
2. Long Cycle Life (up to 1 million cycles)
3. Fast Charging/Discharging (seconds to minutes)
4. Low Self-Discharge
5. High Efficiency (95-98%)
6. Environmentally Friendly
7. Low Maintenance

Disadvantages:

1. Limited Energy Density
2. Lower Voltage Rating
3. Higher Cost per Unit Energy
4. Temperature Sensitivity
5. Limited Depth of Discharge

Applications:

1. Electric Vehicles (regenerative braking)
2. Renewable Energy Systems (smoothing power output)
3. Power Quality and UPS Systems
4. Consumer Electronics (mobile devices, audio systems)
5. Industrial Power Systems (motor starting, backup power)
6. Medical Devices (implantable devices, medical implants)
7. Aerospace and Defense

Difference between Supercapacitors and battery

Sr. No.	Supercapacitors	Battery
1	Store energy electrostatically	Store energy chemically
2	Higher power density	Lower power density
3	Charge/discharge rate is rapid (seconds to minutes),	Charge/discharge rate is longer (minutes to hours).
4	Self-Discharge Rate is slower	Self-Discharge Rate is higher
5	Cycle Life is more	Cycle Life is low
6	Higher cost per unit energy	Lower cost per unit energy
7	Low maintenance	High maintenance
8	Lower voltage rating	Higher voltage rating

Hydrogen Fuel:

Definition:
Hydrogen fuel is a clean-burning fuel that stores energy in the form of hydrogen gas (H2). It can be produced from various sources, including:

1. Fossil fuels (natural gas, coal)
2. Renewable energy sources (solar, wind, hydro power)
3. Biomass
4. Electrolysis of water

Production Methods

1. Electrolysis: Splitting water (H2O) into H2 and O2 using electricity from renewable sources (e.g., solar, wind).
2. Steam Methane Reforming: Converting natural gas (CH4) into H2 and CO2.
3. Biomass Gasification: Converting organic matter into H2-rich gas.
4. Alkaline Electrolysis: Using potassium hydroxide (KOH) electrolyte.
5. Proton Exchange Membrane (PEM) Electrolysis: Using polymer electrolyte.
6. Solar-Powered Electrolysis: Using photovoltaic cells to generate electricity.
7. Biological Production: Microorganisms produce H2 through fermentation.

Storage Methods

1. Compressed Hydrogen: Storing H2 under high pressure (up to 700 bar).
2. Liquid Hydrogen: Cooling H2 to -253°C for storage.
3. Hydrogen Absorption: Storing H2 in metal hydrides or carbon-based materials.
4. Underground Storage: Storing H2 in depleted gas fields or salt caverns.
5. Hydrogen-Loaded Metal Organic Frameworks (MOFs): High-capacity storage.

Advantages:

1. Zero greenhouse gas emissions during combustion
2. Abundant resource (most abundant element in universe)
3. High energy density (120 MJ/kg)

4. Flexible fuel (can be used in various applications)
5. Improves air quality (reduces NOx, SOx, PM)
6. Enhances energy security (reduces dependence on fossil fuels)
7. Supports renewable energy integration

Disadvantages:

1. High production costs
2. Limited infrastructure (hydrogen fueling stations)
3. Storage challenges (high-pressure tanks, liquefaction)
4. Transportation difficulties (hydrogen embrittlement)
5. Safety concerns (flammability, explosion risk)
6. Energy efficiency losses (production, storage, transportation)
7. Competing with battery electric vehicles

Applications:

1. Transportation: Fuel cell electric vehicles (FCEVs), buses, trucks, and aircraft.
2. Power Generation: Hydrogen-fired power plants, fuel cells, and combined heat and power (CHP) systems.
3. Industrial Processes: Chemical synthesis, refining, and manufacturing.
4. Residential Heating: Hydrogen-powered boilers and fuel cells.
5. Maritime Transport: Hydrogen-powered ships and ferries.
6. Aerospace: Hydrogen-powered propulsion systems.
7. Energy Storage: Hydrogen-based energy storage solutions.